1/3 is just the square of the amplitude of that branch—exactly as the Born rule would predict.

There are a couple of lingering worries here. You may object that we considered an especially simple example, where one probability was exactly twice the other one. But the same strategy works whenever we can subdivide our states into the right number of terms so that all of the amplitudes are equal in magnitude. That works whenever the amplitudes squared are all rational numbers (one integer divided by another one), and the answer is the same: probability equals amplitude squared. There are plenty of irrational numbers out there, but as a physicist if you're able to prove that something works for all rational numbers, you hand the problem to a mathematician, mumble something about "continuity," and declare that your work here is done.

We can see Pythagoras's theorem at work. It's the reason why a branch that is bigger than another branch by the square root of two can split into two branches of equal size to the other one. That's why the hard part isn't deriving the actual formula, it's providing a solid grounding for what probability means in a deterministic theory. Here we've explored one possible answer: it comes from the credences we have for being on different branches of the wave function immediately after the wave function branches.

You might worry, "But I want to know what the probability of getting a result will be even before I do the measurement, not just afterward. Before the branching, there's no uncertainty about anything—you've already told me it's not right to wonder which branch I'm going to end up on. So how do I talk about probabilities before the measurement is made?"

Never fear. You're right, imaginary interlocutor, it makes no sense to worry about which branch you'll end up on. Rather, we know with certainty that there will be two descendants of your present state, and each of them will be on a different branch. They will be identical, and they'll be uncertain as to which branch they're on, and they should assign credences given by the Born rule. But that means that all of your descendants will be in exactly the same epistemic position, assigning Born-rule probabilities. So